

Farshad Rahimi Ghashghaei

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🌐 [farshadrahimighashghaei.github.io](https://github.com/farshadrahimighashghaei) 📍 Khuzestan, Iran

Profile

Highly motivated researcher in quantum information science with multiple peer-reviewed publications and international conference presentations. Experienced in cross-institutional collaborations and peer reviewing, contributing to high-impact interdisciplinary projects.

Education

M.Sc. in Cyber Security, Birmingham City University *Jan 2023 – Jan 2024*

- **Thesis:** Post-Quantum Authenticated-Encryption Schemes for QKD
- **Supervisor:** Dr. Mehdi Yousefi
- **Grade:** Distinction (GPA: 4.0/4.0)

B.Eng. in Computer Engineering, Institute for Higher Education ACECR Khuzestan *Sep 2018 – Sep 2022*

- **Supervisor:** Dr. Jamshid Afshani
- **Grade:** 13.82/20 (achieved 19/20 for the final project)

Research Experience

Research Assistant, Birmingham City University *Sep 2023 – Jan 2025*

- Developed post-quantum cryptographic protocols to secure classical and quantum communications.
- Researched quantum control, error correction, and multi-particle modeling.
- Applied machine learning and ensemble methods for quantum simulations and network security.

Publications

Journal papers:

- F. Rahimi Ghashghaei, N. Elmrabit, A.-U.-H. Qureshi, A. Akhunzada, and M. Yousefi, "Advanced Quantum Control With Ensemble Reinforcement Learning: A Case Study on the XY Spin Chain," IEEE Access, vol. 13, pp. 49514–49526, Jan. 2025. [doi:10.1109/ACCESS.2025.3551232](https://doi.org/10.1109/ACCESS.2025.3551232)
- F. Rahimi Ghashghaei, Y. Ahmed, N. Elmrabit, and M. Yousefi, "Enhancing the Security of Classical Communication with Post-Quantum Authenticated-Encryption Schemes for the Quantum Key Distribution," Computers, vol. 13, no. 7, pp. 163, Jul. 2024. [doi:10.3390/computers13070163](https://doi.org/10.3390/computers13070163)
- H. J. Jin, F. Rahimi Ghashghaei, N. Elmrabit, Y. Ahmed, and M. Yousefi, "Enhancing Sniffing Detection in IoT Home Wi-Fi Networks: An Ensemble Learning Approach with Network Monitoring System (NMS)," IEEE Access, vol. 12, pp. 86840–86853, Jun. 2024. [doi:10.1109/ACCESS.2024.3416095](https://doi.org/10.1109/ACCESS.2024.3416095)

Conference papers:

- F. Rahimi Ghashghaei, "Recent Advances in Quantum Error Correction and Their Impact on the Stability of Quantum Computing," 9th International Conference on Interdisciplinary Studies in Nanotechnology, Iran, 2026
- F. Rahimi Ghashghaei, "Prediction of the Behavior of Many-Particle Systems in Quantum Mechanics Using Quantum Neural Networks," 13th International Conference on Applied Research in Basic Sciences, Engineering and Technology, Georgia, 2026

Awards & Honors

- Editor's Choice Article, Computers, 2024 (selected by editorial board).
- MSc with Distinction; among top-ranked students in cohort.

Teaching Experience

Workshop Instructor, Dana Mandegar Pars Institute

Feb 2025 – Jul 2025

- Conducted two workshops on improving student motivation and time management (24-hour each).
- Assessed participant progress and used feedback to refine workshops.

Work Experience

System Administrator, Arya Ravesh Mahshahr Company

Nov 2020 – Dec 2022

- Maintained and monitored servers, including configurations and user access.
- Trained staff on best practices for cyber security and efficient use of resources.

Academic Service

Peer Reviewer for Journals

Dec 2024 – Present

- **Quantum Reports**: 1 review ([Certificate](#))
- **Energies**: 1 review ([Certificate](#))
- **Journal of Computer Sciences and Informatics**: 1 review (certificate available upon request)
- **Symmetry**: 2 reviews ([Certificate 1](#), [Certificate 2](#))

Technical Skills

🖥️ **Quantum Computing & Simulation**: Designed and simulated quantum circuits, implemented Grover, Shor, VQE, QAOA algorithms, modeled noise and decoherence

⚙️ **Quantum Information Theory**: Entanglement measures (entropy, concurrence, negativity), quantum error correction (Shor, Steane, surface codes), quantum communication protocols (BB84, teleportation)

📊 **Machine Learning and Optimization**: Classical ML (regression, classification, clustering), reinforcement learning (Q-learning, DQN, PPO), quantum ML (variational circuits, quantum kernels, hybrid models), with gradient-based optimization and probabilistic modeling experience.

🔧 **Programming and Tooling**: Python (NumPy, SciPy, Pandas, Matplotlib, Qiskit, PennyLane, TensorFlow, PyTorch), C++, Linux, Git, $\text{\LaTeX}$.

Languages

• English – C2 (Advanced) • German – B1 (Intermediate) • Spanish – A1 (Beginner) • Persian – Native

Referees

Available upon request.